

Model studies of the influence of O₂ photodissociation parameterizations in the Schumann-Runge bands on ozone related photolysis in the upper atmosphere

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Abstract. A new parameterization for atmospheric transmission and O₂ photodissociation in the Schumann-Runge band region has been developed and tested with a 1D radiative-photochemical model. The parameterization is based on the O₂-column along the line of sight to the Sun and the local temperature. Line-by-line calculations have served as a benchmark for testing this method and several other, commonly used, parameterizations. The comparisons suggest that differences between the line-by-line calculations and currently accepted parameterizations can be reduced significantly by using the new method, particularly at large solar zenith angles. The production rate of O-atoms computed with this method shows less than 6% deviation compared to the line-by-line calculations at any altitude, all solar zenith angles and in all seasons. The largest errors are found toward the shorter wavelengths in the Schumann-Runge region at low altitudes. Transmittance is approximated to better than 4% at any altitude and/or solar zenith angle. The total O-production rate above 20 km is approximated to better than 2%. The new parameterization is easily implemented in existing photochemical models and in many cases it may simply replace the existing algorithm. The computational effort exceeds that of other parameterizations but in view of the total computation time needed for the actual calculation of the parameterized Schumann-Runge bands this should not lead to significant performance degeneration. The first 14 coefficients of the parameterization are included in this study. Both the complete sets of coefficients and a simple algorithm can be obtained by contacting the authors. A photochemical model study shows the largest effect of the parameterization method is on odd hydrogen concentrations. Subsequent interaction with an odd oxygen family causes differences in the ozone concentrations between the different parameterizations

of more than 10% at selected altitudes. Although it is already established that deficiencies in the treatment of Schumann-Runge band absorption are unlikely to explain the current underestimation of ozone concentration at the stratopause in a variety of photochemical models, this study does show that the choice of parameterization has a large impact on the accuracy of the results at large solar zenith angles and in different seasons.

1 Introduction

Absorption of UV solar radiation in the mesosphere and stratosphere is dominated by molecular oxygen and ozone. Absorption by oxygen between 137–242 nm, partitioned by the dominating molecular transitions of O₂, can be divided into three different wavelength intervals: the Schumann-Runge continuum (137–175 nm), the Schumann-Runge bands, $X^3\Sigma_g^- \rightarrow B^3\Sigma_u^-$ (175–205 nm), and the Herzberg continuum, $X^3\Sigma_g^- \rightarrow A^3\Sigma_u^+$ (185–242 nm).

Due to the high degree of predissociation of the $B^3\Sigma_u^-$ state the absorption of a photon with a wavelength within the Schumann-Runge (SR) bands results in photolysis leading to two oxygen atoms in the ground state. Photodissociation in the SR bands is the main source of odd-oxygen in the mesosphere and remains important throughout the stratosphere. Figure 1 shows a typical profile of O-production rates partitioned by the wavelength intervals as defined above, for overhead Sun and component distribution by the US Standard Atmosphere (1976). Below about 60 km the Herzberg continuum dominates and above 90 km the Schumann-Runge continuum is the major contributor. Both the Schumann-Runge and the Herzberg continua contribute significantly to oxygen photolysis at the short and long wavelength end of the SR band region, respectively.

Lyman- α at 121.6 nm is able to penetrate to about 60 km above which it contributes significantly to the

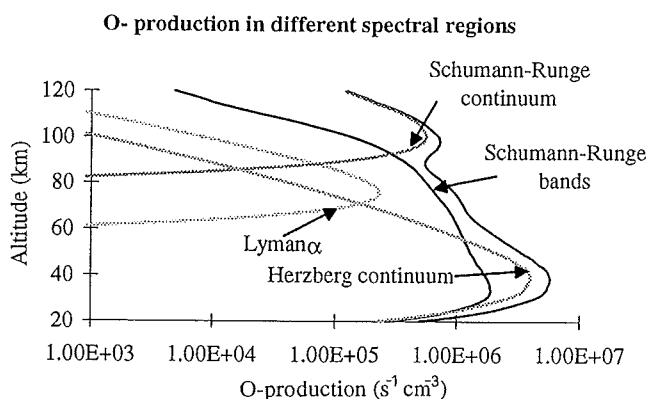


Fig. 1. A comparison of the O-production from O₂ photodissociation in the spectral regions of the Schumann-Runge bands, the Schumann-Runge continuum, the Herzberg continuum and Ly- α as a function of altitude (see text for the corresponding wavelengths). The profiles are obtained from a model run with overhead Sun US Standard Atmosphere (1976) and WMO 1985 values for the Ly- α flux

O-production rate. Moreover from the point of view of mesospheric and stratospheric chemistry the Lyman- α line is important since about 70% of the H₂O photodissociation above 70 km occurs at this wavelength. Generally the Ly- α line is included separately in photochemical models. Figure 1 indicates that in the lower stratosphere, below 25 km, the Schumann-Runge bands spectral region (175–205 nm) for these particular conditions again becomes the dominant O source rather than of photodissociation at longer wavelengths.

The existence of strong lines in the SR-band region and a “quasi-continuum” between them implies that solar radiation can penetrate into the stratosphere at discrete wavelengths where it contributes significantly to dissociation. Since the line shapes, and through them the penetration depth, are strongly dependent on atmospheric variables such as temperature and pressure, solar transmittance is extremely sensitive to the variation of these variables along the optical path. Optical paths are generally not homogeneous in the atmosphere and vary with different solar elevations (variations along the line of sight to the Sun) and different seasons (temperature, density variations). An algorithm to calculate dissociation rates and transmittance based on solely the O₂ column density on the line of sight to the Sun (hereafter called: slant O₂ column density) is therefore not likely to perform well under all circumstances.

An accurate determination of solar irradiance, or atmospheric transmittance, is not only important to determine atomic oxygen production rates, but may also affect photolysis of trace gases. These species that do not substantially modify the solar irradiance due to their low concentrations can influence chemistry through subsequent chemical reactions. In view of this fact it is extremely important to calculate accurately both atmospheric transmittance and photodissociation rates in this wavelength region. In the Schumann-Runge bands this would mean resolving more than 7000 individual spectral lines at each grid point of the photochemical model;

Table 1. Interval labels and spectral regions

Region number	Wave number range (cm ⁻¹)	Wavelength range (nm)
1	57,000–56,500	174.4–177.0
2	56,500–56,000	177.0–178.6
3	56,000–55,500	178.6–180.2
4	55,500–55,000	180.2–181.8
5	55,000–54,500	181.8–183.5
6	54,500–54,000	183.5–185.3
7	54,000–53,500	185.2–186.9
8	53,500–53,000	186.9–188.7
9	53,000–52,500	188.7–190.5
10	52,500–52,000	190.5–192.3
11	52,000–51,500	192.3–194.2
12	51,500–51,000	194.2–196.1
13	51,000–50,500	196.1–198.0
14	50,500–50,000	198.0–200.0
15	50,000–49,500	200.0–202.0
16	49,500–49,000	202.0–204.1
17	49,000–48,500	204.4–205.8

imposing a calculation volume that is excessive even with current computer speeds. A parameterization should ideally increase computational speed as much as possible without sacrificing the accuracy of a detailed calculation.

The choice of independent variables in such a parameterization will prove to be essential for the ability of the parameterization to reproduce direct calculations with a high spectral resolution algorithm. One must evaluate the balance between the required accuracy, against the resulting computational speed. Traditionally, an often adopted approach to reduce the computational complexity is to subdivide the Schumann-Runge band wavelength region into the 17 spectral intervals denoted in Table 1 (WMO, 1985) and to parameterize averaged transmittance and photodissociation rates in each interval. Three parameterization methods that employ these 17 spectral intervals, all using different independent variables, are commonly used and will be compared here. Allen and Frederick (1982) parameterize effective cross sections as a function of slant oxygen column density and, depending on the interval, temperature or pressure. From the effective cross section one can calculate both transmittance and photodissociation rates. WMO (1985) provides a reference table for transmittance at different oxygen columns from which dissociation rates can be calculated. In view of the previous discussion this is certainly not adequate when high accuracy is required. Nicolet and Kennes (1989a) use slant oxygen column density and solar zenith angle to parameterize dissociation rates, transmittance (and/or optical depth) separately. In this study we present a parameterization of the effective cross section that is based on the slant oxygen column density and the local temperature.

Recently a new parameterization has become available (Kockarts, 1994), based on reduction functions. Here a polynomial fit of oxygen absorption cross section as function of wavelength and temperature (Minschwaner *et al.*, 1992) has been used in a two-parameter approach.

2 Parameterization in photochemical models

2.1 Introduction

Generally, the calculations of oxygen photodissociation in atmospheric models in the wavelength region 137–242 nm involves calculation of both local transmittance and photodissociation rates according to:

$$T(\lambda, z) = \exp\left(-\sum_i \int_z^\infty \sigma_i(\lambda, \theta_z) n_i dl\right) \quad (1)$$

and

$$J_{O_2}(\lambda, z, \theta_z) = \sigma_{O_2}(\lambda, \theta_z) T(\lambda, z) I_\infty(\lambda). \quad (2)$$

Since the absorption cross section σ in the atmosphere generally depends on the local temperature θ_z , then the transmittance T at an altitude z and at wavelength λ depends on the thermal history along the line of sight to the Sun as in Eq. (1). The summation over i considers all sources of attenuation of solar radiation at the given wavelength. In the Schumann-Runge band region these are oxygen and ozone absorption.

$$T = T_{O_3} T_{O_2}. \quad (3)$$

Equation (2) defines the rate J_{O_2} at which oxygen is photodissociated at a wavelength λ . Clearly it depends on the absorption cross section σ_{O_2} , the transmittance given by Eq. (1) and the radiative flux at the top of the atmosphere. J_{O_2} depends therefore in principle on both the local temperature and the temperature history along the optical path. Global-scale photochemical models, but even 1D and 2D models, require fast algorithms to compute transmittance and photodissociation rates at all altitude levels in the model. Equations (1) and (2) are only valid for wavelength intervals over which σ does not vary significantly. The Schumann-Runge bands, however, show significant spectral structure over small wavelength intervals and suitable spectral averaging is required to reduce the complexity. A parameterization of the Schumann-Runge bands must yield spectrally averaged transmittance and oxygen photodissociation rates. Equations (1) and (2) indicate that this can be achieved by parameterizing the transmittance and, either the photodissociation rate or the absorption cross section leading to the two-parameter approach mentioned above. However, a single-parameter approach involving just the absorption cross section will prove to be sufficient to recover both spectrally averaged transmittance and photodissociation rates.

2.2 Spectral averaging of the absorption cross section

With the rapidly varying cross sections in the SR region it is expedient to define an effective cross section σ_{eff} and an averaged transmittance $T_{SR}(\Delta\lambda)$ due to oxygen opacity in a wavelength interval $\Delta\lambda$ as:

$$\sigma_{eff}(\Delta\lambda, z, \theta_z) = \frac{\int \sigma_{O_2}(\lambda, \theta_z) I_\infty(\lambda) T_{O_2}(\lambda, z) d\lambda}{\int I_\infty(\lambda) T_{O_2}(\lambda, z) d\lambda}, \quad (4a)$$

$$T_{SR}(\Delta\lambda, z) = \frac{\int I_\infty(\lambda) T_{O_2}(\lambda, z) d\lambda}{\int I_\infty(\lambda) d\lambda}. \quad (4b)$$

The effective cross section σ_{eff} as defined is the average cross-section in a spectral interval where the actual cross sections are weighted by the local irradiance. This quantity is valid at a given altitude z or oxygen column density N_{O_2} and depends on the temperature along the optical path. By differentiating the natural logarithm of Eq. 4b with respect to N_{O_2} and Lambert-Beer's law it follows that the two quantities are interrelated as:

$$\frac{d \ln T_{SR}}{d N_{O_2}} = -\sigma_{eff}. \quad (5a)$$

The spectrally averaged oxygen photodissociation $J_{O_2}(\Delta\lambda)$ in a spectral interval $\Delta\lambda$ at an altitude z can be defined as:

$$J_{O_2}(\Delta\lambda) = \sigma_{eff}(\Delta\lambda) T(\Delta\lambda) I_\infty(\Delta\lambda), \quad (5b)$$

where $T(\Delta\lambda)$ is defined as the spectral average from Eq. (3) and the spectral averaged solar irradiance is: $I_\infty(\Delta\lambda) = (1/\Delta\lambda) \int I_\infty(\lambda) d\lambda$. In an isothermal atmosphere σ_{eff} decreases with altitude in spectral intervals with strong absorption features since radiation near the line centres is absorbed at higher levels. This is clearly demonstrated in Fig. 2 where the effective cross sections are plotted against slant O₂ column density to reduce the solar zenith angle effect. Two intervals are included to demonstrate the general features in a non-isothermal

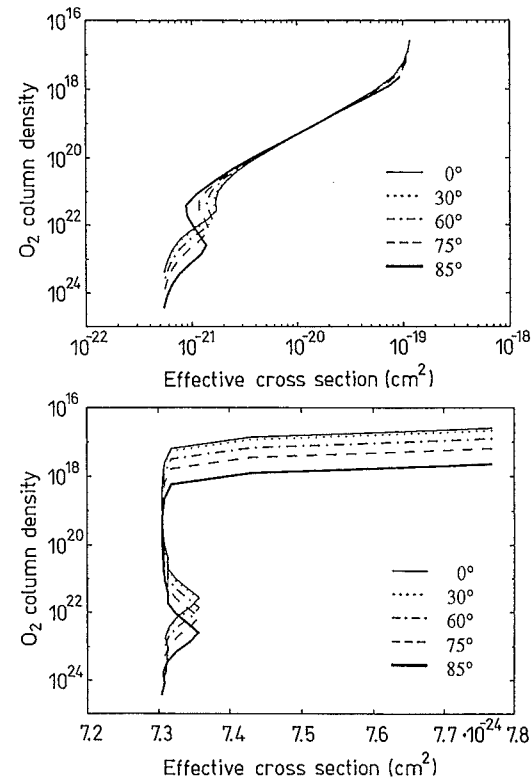


Fig. 2. Effective cross section in interval 2 (56500–56000 cm⁻¹) and interval 17 (49000–48500 cm⁻¹) for five solar zenith angles: 0° (—), 30° (···), 60° (---), 75° (---) and 85° (—) as function of the O₂ column density

atmosphere. Several properties may be noticed: (1) within all intervals considerable differences in the effective cross sections at a given O₂ column density are observed for different solar zenith angles. This indicates that a simple parameterization in terms of only O₂ column density will lead to significant errors. (2) The effective cross section in interval 2, at the short-wavelength end (see Table 1), where the absorption cross section has strong absorption lines, shows significantly increased values at column densities around the warm stratopause and in the thermosphere. This is due to the emergence of significant absorption from the thermally excited $v'' = 1$ state that has a significant contribution to the effective cross section in intervals 1 to 4. In the other wavelength intervals a less significant increase of the effective cross sections is also observed. (3) At low altitudes the cross section approximates its quasi-continuum value since insolation at the strong absorption features will have been absorbed at higher altitudes and no longer contributes to the integral in Eq. (4a). The effective cross sections in intervals 16 and 17, at long wavelengths where weaker absorption features are observed do not show much altitude variation. Some parameterization therefore do not include the two longest wavelength intervals but choose to treat these intervals with constant absorption cross sections.

It should be noted that the effective absorption cross section σ_{eff} is the relevant quantity when calculating the photodissociation rate [Eq. (5b)]. The effective absorption cross section, which is a local, spectrally weighted quantity should not be confused with the quantity $\sigma = -\ln(T)/N$ that is spatially averaged absorption cross section (it may or may not be spectrally averaged) along the optical path from the top of the atmosphere. This distinction is essential, particularly when using transmittance tables such as given by (WMO, 1985). In such a case the photodissociation rate might be obtained from the transmittance when the local effective absorption cross section is calculated using Eq. (5a). Using the spatially averaged absorption cross section according to the definition in this paragraph with Eq. (5b) to calculate photodissociation rates will lead to erroneous results.

Knowledge of σ_{eff} gives, by application of Eqs. (3)–(5a), all the necessary information to determine both the atmospheric transmittance and the oxygen photodissociation rates in the Schumann-Runge band region. Unsatisfactory results using only σ_{eff} have been reported by Nicolet and Kennes (1989b). In the next paragraph we will show that by applying a new altitude-averaging scheme the line-by-line calculations may be well approximated using the effective cross section as a single parameter.

2.3 Calculating the transmittance from σ_{eff}

Since, depending on the spectral interval, σ_{eff} changes by several orders of magnitude from 150 km to 20 km, we find that, when evaluating the difference approximation of the integral form of Eq. (5a) at discrete model levels with $\Delta z > 1$ km (which is often the case in photochemical models), large errors in T_{SR} can occur. To avoid this it is necessary to account for the change in σ_{eff} between two

adjacent layers. Using an average effective cross section at the top and bottom of a layer of the form $\ln(\Delta T) = -\frac{1}{2}(\sigma_z + \sigma_{z+\Delta z})(N_{O_2,z} - N_{O_2,z+\Delta z})$ as used, for example by Allen and Frederick (1982) is not sufficient, although quite common. We present another approach, necessary to obtain correct transmittance when using these kinds of parameterizations.

Figure 2 shows that the variation of $\ln(\sigma_{eff})$ with $\ln(N_{O_2})$ in small height (or oxygen column density) intervals the relationship is nearly linear. We therefore choose the following approximation for σ_{eff} within a layer:

$$\ln(\sigma_{eff}(N_{O_2})) = c_1 \cdot \ln(N_{O_2}) + c_0, \quad (6)$$

where the constants c_1 and c_0 can be expressed in the effective cross sections at the top and bottom of the layer. In fact, this approach assumes an exponential variation of both σ_{eff} and N_{O_2} with altitude within one layer. The natural logarithm of the transmittance ΔT_{SR} through a layer between the discrete levels N_{O_2} and $N_{O_2} + \Delta N_{O_2}$ is then:

$$\ln(\Delta T_{SR}) = - \int_{N_{O_2}}^{N_{O_2} + \Delta N_{O_2}} \exp(c_1 \ln(N) + c_0) dN. \quad (7)$$

Substituting for c_0 and c_1 and evaluating this integral T_{SR} can be expressed in quantities at the top and bottom of the layer:

$$\ln(\Delta T_{SR}) = \frac{\sigma_{eff} \cdot N - \sigma_{eff,N+\Delta N} \cdot [N + \Delta N]}{1 + \ln\left(\frac{\sigma_{eff,N}}{\sigma_{eff,N+\Delta N}}\right) \bigg/ \ln\left(\frac{N}{N + \Delta N}\right)}, \quad (8)$$

where we omit the O₂ subscript for readability. The transmittance T_{SR} from the top of the atmosphere to an altitude z in a wavelength interval in the Schumann-Runge bands can then be calculated by traversing from the top of the atmosphere down to z calculating the transmittance in each layer with Eq. (8) and:

$$T_{SR} = \prod \Delta T_{SR}. \quad (9)$$

There are no restrictions in the use of Eq. (8) on the slope of the effective cross section (e.g. increasing effective cross sections with decreasing altitude and the stratopause). As long as the effective cross section is not discontinuous function of the column density Eq. (8) will reduce to $\ln(\Delta T_{SR}) = -\sigma_{eff} \Delta N$ when either $\Delta \sigma_{eff}$ or ΔN approaches zero.

3 Calculations

As described in detail by Murtagh (1988) and (1989) we subdivide the Schumann-Runge bands into 17 spectral intervals of 500 cm⁻¹ width following (WMO, 1985) standard practice. Absorption cross sections are calculated at full resolution (0.02 cm⁻¹). Lines from bands with $v' = 0-19$, $v'' = 0-1$ and $N'' = 1, 3, 5, \dots, 31$ are included. The calculation is done at four temperatures (150, 200, 250 and 300 K) to cover stratospheric and mesospheric temperatures. The resulting cross sections include

both the Schumann-Runge and Herzberg continua, the latter now modified to include the results of Yoshino *et al.* (1988). Care has been taken to match the referenced Herzberg continua data to our line-by-line calculations.

3.1 The parameterization method

Using the US Standard Atmosphere (1976) atmospheric transmittance T is calculated at full spectral resolution following the absorption of solar radiation with Eq. (1) for 5 different solar angles between overhead Sun and 5 degrees above the horizon. At each level σ_{eff} was then calculated for all 17 spectral intervals using Eq. (4a). Transmittance and dissociation rates [Eq. (2)] are also calculated at full resolution for typical high-latitude summer and winter atmospheres. These will act as test cases for the performance of the parameterization under different atmospheric conditions.

While seeking a suitable method to parameterize the effective cross section, inspection showed a near linear relationship between the logarithm of σ_{eff} and the local temperature at a fixed O₂ column density. Figure 3 illustrates this by plotting the effective cross section in one interval at a fixed oxygen column density, obtained from the calculations at the five solar zenith angles, against local temperature. Accordingly, for a given O₂ column density, the effective cross section can be expressed as:

$$\ln(\sigma_{eff}(N_{O_2})) = A(N_{O_2})[\theta(N_{O_2}) - 220] + B(N_{O_2}), \quad (10)$$

where N_{O_2} is the slant column density along the absorbing path and $\theta(N_{O_2})$ the local temperature in Kelvin at the corresponding altitude in the atmosphere. Parameters A and B were calculated in each interval for slant O₂ column densities between $38 < \ln(N_{O_2}) < 56$. B can be considered as representing the effective cross section at 220 K and A gives the temperature dependency. They correspond to slope and the intercept at 220 K in Fig. 3. To confirm that the linear relationship is general applicable Fig. 4 illustrates some ranges between 0 and 1 for the correlation coefficient r^2 for the 17 spectral intervals at all column densities. In 72% of the cases the regression coefficient r^2 is better than 0.95 and in 90% of all cases better

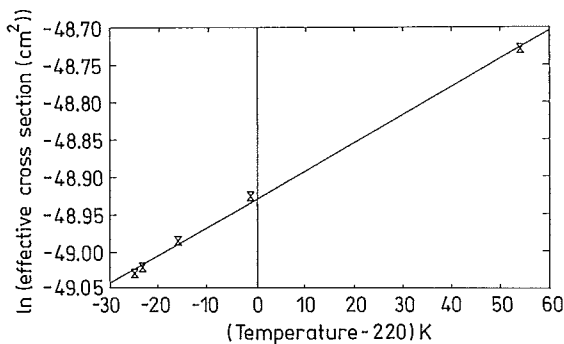


Fig. 3. The variation of the effective cross section with the local temperature for a slant O₂ column density of 10^{18} cm^{-2} in interval 10

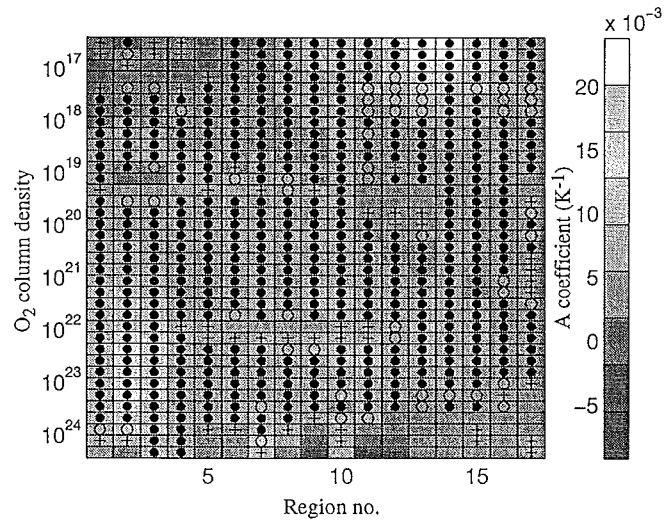


Fig. 4. Regression coefficient r^2 of the linear fit and the A coefficient according to Eq. (10). Three intervals are indicated for r^2 : > 0.95 (●); $0.9 < r^2 < 0.95$ (○); $0.7 < r^2 < 0.9$ (+)

than 0.70, indicating an excellent fit. Those points in Fig. 4 with an $r^2 < 0.7$ are limited to regions that are optically thick for the spectral interval considered (e.g. high column densities) or spatial regions where the effective cross section has a very low local temperature dependence (equivalent to a horizontal slope in a plot similar to Fig. 3). The altitude region where this occurs is actually exactly under a temperature maximum. Since “hot” lines have been present in the absorption history, their presence in reduced strength, because of the lower temperature, does not lead to a strong local temperature dependence, unless the O₂ amount is significantly increased. The latter indicates that at some altitude below the temperature maximum a stronger dependency on local temperature should return, which can be confirmed in Fig. 4.

As shown by Kockarts (1976) the following approximation for the absorption cross section at a given wavelength for a single line in a band for the O₂ molecule can be given as:

$$\sigma(\theta) = \frac{\alpha}{\theta} \exp\left(-\frac{\beta N''(N'' + 1)}{\theta}\right), \quad (11)$$

where α and β are constants independent of the temperature θ . The absorption cross section can either increase or decrease, depending on the rotational quantum number N'' . The functional dependency of the effective absorption cross section (a spatial and spectral average) on temperature is, of course, more complicated than indicated by Eq. (11). As described above, the actual dependency is, however, to good approximation, linear in local temperature which is not inconsistent with a Taylor expansion of Eq. (11) around 220 K with errors of the order $O((\theta/220 - 1)^2)$.

Figure 5 shows the variation of both parameters with O₂ column density for interval 10. As can be seen, both parameters have reasonably smooth variation with $\ln(N_{O_2})$ and should be suitable for polynomial expansion.

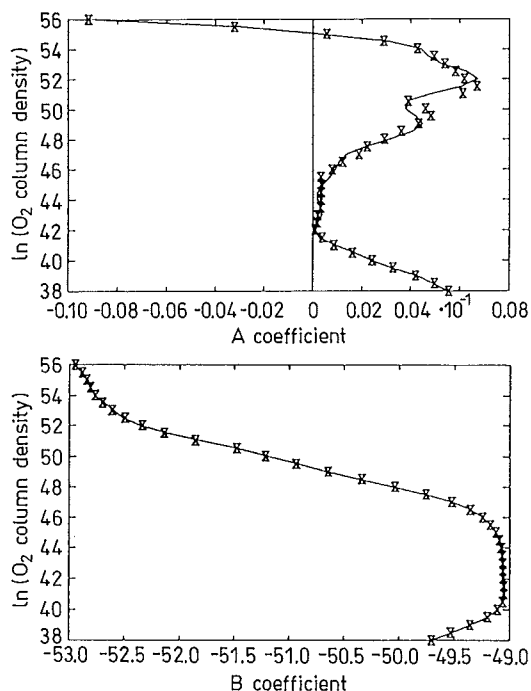


Fig. 5. Variation of the parameters A and B with O₂ column density in interval 10. The Chebeshev polynomial fit evaluated with the first 12 terms is included to illustrate the fitting accuracy

The corresponding curves for each spectral interval were fitted as a function of $\ln(N_{O_2})$ using a Chebeshev polynomial expansion of the order of 20. The reason for choosing Chebeshev polynomials was that this type of expansion is easily truncated at any desired number of terms, without having to refit and at the same time minimizing the maximum deviation. Visual inspection of the fitted curves indicates that no more than 10 or 12 terms are needed to obtain an excellent fit. This has been verified by calculating maximum errors induced by the fitting procedure for different number of terms. In Fig. 6 the total maximum error between the parameterization and the line-by-line calculations are presented as a function of the number of polynomial coefficients used in the parameterization. The figure shows that no significant improvement is obtained using more than approximately 12 terms. Table 2 includes the first 14 Chebeshev polynomial coefficients for the parameters A and B . All 20 coefficients can be obtained by contacting the authors or pointing your web browser at http://www.misu.su.se/srb_param.html. For O₂ column densities between $38 < \ln(N_{O_2}) < 56$, these can be used to retrieve the A and B parameters used in Eq. (10). The chosen limits can be converted to optical depths, given typical extremes in the cross section (e.g. $5e-24$ and $1e-19$ for minimum and maximum, respectively). The upper boundary then corresponds to an optical depth of approximately 0.003 for the strongest S/R absorption feature, and similarly the lower boundary to correspond to an optical depth of more than 10 for the weakest absorption feature, justifying these limits. Transmittance and photodissociation rates at lower altitudes can be obtained by using values for the effective cross

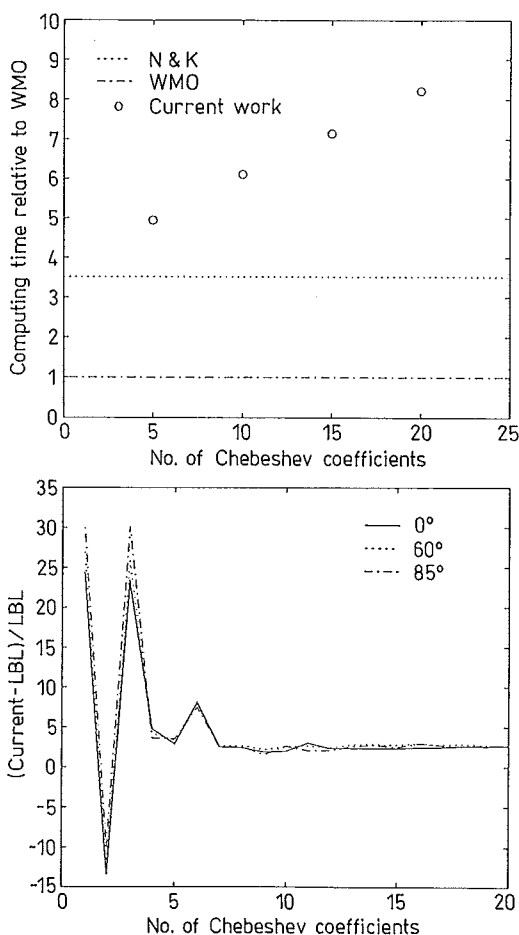


Fig. 6. Parameterization induced error and computational speed as a function of the number of coefficients in the Chebeshev expansion. Error is given in percent and computational speed is compared to the WMO transmittance reference table parameterization

section from the lowest valid altitude range. This is justified since at these altitudes all strong absorption features have become optically thick, implying a nearly constant effective cross section with altitude. This can be verified in Fig. 2.

3.2 Photochemical modelling

To illustrate the effect of the current work and to estimate the sensitivity of ozone chemistry on parameterization methods in the Schumann-Runge band region a 1D radiative diffusive model has been used. The basics of such models are discussed extensively in the literature (e.g. Rodrigo *et al.*, 1986; Battaner, 1975). The model is intended to describe the atmosphere between 40 and 150 km. We specifically address the consequences of the parameterization presented here in a simple oxygen hydrogen atmosphere and the question whether it is worth implementing it despite a small increase in computational effort compared to the WMO scheme. Recently, Siskind *et al.* (1994) made a similar comparison between A & F (Allen and Frederick, 1982), WMO and their own parameterization. Seasonal effects were not addressed.

Table 2a. The first 14 Chebyshev polynomial coefficients for the parameter A

Region	c ₀	c ₁	c ₂	c ₃	c ₄	c ₅	c ₆	c ₇	c ₈	c ₉	c ₁₀	c ₁₁	c ₁₂	c ₁₃
1	1.0329E-2	5.1324E-3	-3.0092E-3	-2.0032E-3	-8.8428E-4	-3.8237E-4	-3.3118E-4	-1.6750E-4	5.2790E-5	5.0426E-4	4.4909E-4	-7.8364E-6	-7.5366E-5	6.5189E-5
2	1.2459E-2	6.6948E-3	-3.6213E-3	-4.5405E-3	-3.0803E-3	-4.7438E-4	-1.2366E-4	-4.0857E-4	-1.6714E-4	1.7378E-4	4.2953E-4	3.5813E-4	-1.9126E-4	-2.1135E-4
3	1.1748E-2	6.7515E-3	-1.7180E-3	-2.4305E-3	-1.6946E-3	1.5557E-4	5.6670E-4	3.6552E-4	-2.0621E-4	-5.5456E-4	4.6761E-5	4.8784E-4	-2.7043E-5	-2.9904E-4
4	6.0013E-3	3.0424E-3	-1.3456E-3	-3.1058E-3	5.4464E-4	2.6087E-6	-7.9800E-5	3.7031E-4	7.0893E-5	-1.9535E-4	-1.4547E-4	-9.3527E-5	8.3172E-5	1.3192E-4
5	3.8316E-3	9.2559E-4	-2.4159E-3	-7.5083E-4	1.2181E-4	-4.5613E-5	-2.6216E-4	1.6983E-5	-4.7811E-5	-1.2092E-4	-1.0777E-5	2.2979E-5	4.9988E-5	8.8607E-5
6	4.1947E-3	-6.8755E-5	-1.4245E-3	-8.6076E-4	2.0853E-4	-2.0373E-4	-4.5861E-4	9.2391E-5	1.4860E-4	-9.6231E-5	-1.6515E-4	-5.4423E-5	1.2346E-4	1.6950E-4
7	8.6832E-3	3.6594E-3	8.1732E-4	6.0567E-4	1.3563E-3	5.9450E-4	-3.0034E-4	-4.6217E-4	-4.7349E-4	-2.4793E-4	-8.9347E-5	-1.0779E-4	1.2359E-4	3.6646E-4
8	6.6173E-3	-2.1543E-4	1.4047E-4	-1.1793E-3	6.1219E-4	-6.0140E-5	-6.3005E-4	-6.6610E-5	-5.1488E-5	-1.0091E-4	-1.7954E-4	-2.0512E-4	1.2008E-4	3.3955E-4
9	5.4550E-3	-3.6226E-3	-1.3652E-3	-3.6097E-3	-6.1758E-4	-4.3579E-4	-8.0372E-4	-2.7473E-4	-1.5659E-4	6.1094E-5	2.6251E-5	-6.4449E-5	1.3693E-4	2.5882E-4
10	9.3783E-3	-1.0028E-3	-3.1705E-4	-6.0869E-4	7.7372E-4	3.8064E-4	5.2628E-4	3.6868E-4	-5.5993E-4	-5.4814E-4	-1.7028E-4	-1.4983E-5	1.3437E-4	2.1617E-4
11	3.7310E-3	-2.1483E-3	-1.2498E-3	-4.2744E-3	-1.6933E-3	-7.9122E-4	-5.6881E-4	-1.3753E-4	-1.7168E-4	2.3004E-5	-3.8134E-5	-1.8368E-4	4.1316E-5	2.4083E-4
12	4.2505E-3	-4.0228E-3	2.5235E-4	-4.0408E-3	-6.1246E-4	-5.3371E-5	-2.0204E-4	1.9077E-5	-4.5119E-4	-1.2529E-4	8.8716E-5	-2.6860E-5	5.8056E-5	1.3005E-4
13	1.1185E-2	-8.1540E-3	5.0569E-3	-5.7041E-3	1.0066E-3	8.5342E-4	-5.1804E-6	5.5014E-4	-7.8335E-4	-2.8769E-4	1.9516E-4	-1.6853E-5	2.8479E-5	4.9984E-5
14	1.2223E-2	-8.4868E-3	4.9675E-3	-4.5895E-3	9.8124E-4	7.1158E-4	-2.3529E-4	5.4889E-4	-4.1330E-4	-2.3418E-4	4.7354E-5	-8.4890E-5	1.3070E-5	8.0553E-5
15	6.5869E-3	-4.3355E-3	2.6363E-3	-2.2173E-3	6.8916E-4	2.2469E-4	-1.2529E-4	3.6422E-4	-2.6622E-4	-1.3002E-4	4.9132E-5	-3.7708E-5	1.9096E-5	2.6320E-5
16	2.1501E-3	-1.6598E-3	1.1854E-3	-9.1335E-4	2.5185E-4	8.0311E-5	-7.8130E-5	1.3029E-4	-1.0238E-4	-4.3029E-5	2.5849E-5	-1.5650E-5	2.6531E-6	9.3739E-6
17	7.1933E-4	-4.1386E-4	4.2634E-4	-2.0867E-4	1.1925E-4	6.3301E-5	-9.1309E-7	4.9583E-5	-2.9524E-5	-2.0359E-5	4.8154E-6	-5.6399E-6	-2.6650E-6	1.9148E-6

Table 2b. The first 14 Chebyshev polynomial coefficients for the parameter B

Region	c ₀	c ₁	c ₂	c ₃	c ₄	c ₅	c ₆	c ₇	c ₈	c ₉	c ₁₀	c ₁₁	c ₁₂	c ₁₃
1	-9.0338E+1	-2.0433E+0	5.5426E-1	2.6353E-1	-1.9082E-1	-8.9445E-2	1.0309E-1	8.6853E-3	-1.9734E-2	-2.7485E-2	1.7318E-2	2.1101E-2	-1.7113E-2	3.3153E-3
2	-9.3233E+1	-3.1658E+0	6.3663E-1	3.9178E-1	-1.4380E-1	-9.3397E-2	1.7969E-2	-1.0547E-2	2.9619E-2	-1.9000E-2	-7.3326E-3	1.3821E-2	-4.4318E-3	4.1003E-4
3	-9.3862E+1	-2.9470E+0	5.7835E-1	3.8779E-1	-1.4584E-1	-7.0764E-2	4.5143E-2	-5.1220E-2	2.6228E-2	1.8199E-2	-1.1982E-2	5.9013E-3	-1.0953E-2	-7.7679E-3
4	-9.4098E+1	-2.9064E+0	3.5364E-1	5.5376E-1	-1.1709E-1	-1.4125E-1	4.8244E-2	-2.3598E-2	2.0288E-2	2.8259E-2	-1.7176E-2	-1.3475E-2	3.6190E-3	3.8261E-3
5	-9.5338E+1	-3.0880E+0	2.7569E-1	5.7639E-1	-7.6243E-2	-1.4208E-1	3.2333E-2	-2.0989E-2	1.7372E-2	1.5909E-2	-1.5132E-2	1.9616E-3	5.2331E-3	-8.208E-3
6	-9.6418E+1	-2.8580E+0	5.8291E-2	6.3749E-1	-5.3117E-2	-1.4818E-1	2.8866E-2	-1.0763E-2	-1.1160E-2	2.9793E-2	8.2370E-3	-1.8322E-2	-2.4516E-3	1.8008E-3
7	-9.7031E+1	-2.5935E+0	-7.0244E-2	6.0010E-1	1.5120E-2	-1.6378E-1	7.4340E-3	1.0988E-2	-2.1901E-2	1.7579E-2	2.6648E-2	-1.2947E-2	-1.8792E-2	-1.8242E-3
8	-9.8644E+1	-2.5357E+0	-3.2940E-1	6.5049E-1	5.2197E-2	-1.0580E-1	-2.7340E-2	1.4917E-2	-1.3663E-3	1.2981E-4	1.7599E-2	-1.0151E-2	-9.0692E-3	1.6539E-3
9	-1.0001E+2	-2.2519E+0	-5.1556E-1	6.2480E-1	6.0292E-2	-8.3070E-2	-6.3975E-2	1.6936E-2	-7.2830E-3	-2.2170E-2	3.6119E-2	1.0714E-2	-1.7677E-2	-1.3045E-2
10	-1.0111E+2	-2.0825E+0	-4.8529E-1	4.8028E-1	1.3507E-1	-6.8641E-2	-5.5579E-2	3.1236E-2	4.5994E-3	-1.7860E-2	2.2736E-2	5.9568E-3	-1.7614E-2	-1.1374E-2
11	-1.0136E+2	-2.0662E+0	-4.7455E-1	4.7455E-1	2.1683E-1	-1.9375E-2	-1.3641E-1	-2.0723E-2	3.5558E-2	-9.5115E-3	5.0735E-3	8.1467E-3	2.5605E-3	-1.2672E-2
12	-1.0296E+2	-1.3375E+0	-7.4955E-1	4.2104E-1	9.0381E-2	5.2656E-2	-1.2852E-1	-2.1576E-2	5.2225E-2	-5.7592E-3	-1.9192E-3	-1.1614E-3	6.1514E-3	5.2872E-2
13	-1.0517E+2	-2.0535E-1	-9.1837E-1	6.4242E-1	-2.0129E-1	1.7411E-1	-1.4023E-1	-1.9981E-2	7.4455E-2	-2.8757E-2	1.8494E-2	-2.4927E-2	1.1849E-3	1.2711E-2
14	-1.0651E+2	3.9179E-1	-6.7713E-1	4.5236E-1	-2.5798E-1	1.8685E-1	-7.4748E-2	-2.6506E-2	3.9407E-2	-2.9446E-2	2.9447E-2	-1.2442E-2	-5.6759E-3	4.0942E-3
15	-1.0644E+2	1.9692E-1	-3.8104E-1	2.4173E-1	-1.3859E-1	1.0406E-1	-3.9505E-2	-1.2371E-2	1.8715E-2	-1.7700E-2	1.7964E-2	-6.4588E-3	-2.9060E-3	2.0088E-3
16	-1.0658E+2	1.1428E-1	-1.4288E-1	9.7194E-2	-6.3228E-2	4.0837E-2	-1.3954E-2	-2.9513E-3	7.9819E-3	-8.5961E-3	6.0861E-3	-2.5232E-3	-3.6467E-4	1.4113E-3
17	-1.0659E+2	4.2359E-2	-3.6721E-2	2.9410E-2	-1.9349E-2	1.0664E-2	-4.2608E-3	-5.0675E-4	2.9290E-3	-2.2630E-3	1.6710E-3	-8.3252E-4	-3.0056E-4	4.2618E-4

The model calculates 17 species including the odd-oxygen family ($O_x = O(^1D) + O(^3P) + O_3$), the odd-hydrogen family ($HO_x = H + OH + HO_2$) and other hydrogen-containing species such as H₂O, H₂O₂, and H₂. The included carbon compounds are CH₄, CO₂ and CO. The set of equations for a chemical species n_i with a production rate P_i and loss rate L_i :

$$\frac{\delta n_i}{\delta t} = \sum_i (P_i - L_i \cdot n_i) + \frac{\delta}{\delta z} \phi_i \quad (12)$$

from a stiff set of partial differential equations (Gear, 1971). ϕ_i denotes the flux of species i due to molecular and eddy diffusion. To reduce the stiffness of the set of Eq. (12) we used the chemical family concept where the groups as a whole are calculated with Eq. (12) and the concentration of individual family members are obtained from steady-state assumptions between the family members. We applied an implicit Crank-Nicholson scheme for the families and species with sufficiently long chemical lifetimes. A total of 31 chemical reactions and 8 photodissociation processes are considered. Boundary conditions are either diffusive equilibrium or constant mixing ratio for all species except for hydrogen and oxygen at the upper boundary.

Although the model may in principle provide full diurnal coverage, we will discuss here two cases at fixed solar zenith angles: namely overhead Sun and 85°. The reason for discussing two steady-state runs instead of the diurnal runs is that the latter gives similar results but the steady-state solutions explain the main features. In the diurnal mode the model is initialized with an atmosphere from a run from one of these steady-state modes of the model. These are in their turn initialized with a typical atmosphere for 60° latitude at the summer solstice. This means that for the diurnal model run the solar zenith angle lies between 30° and 90° between noon and sunset/sunrise. This highlights the parameterization performance at high zenith angles. The diurnal calculations are performed at constant day number, so that photodissociation rates remain constant with respect to the time of the day, until a stable solution is obtained.

4 Results

4.1 Line-by-line and parameterization results

To evaluate the direct errors induced by using a parameterization instead of a high-resolution line-by-line code, we compared three parameterization methods: Allen and Frederick (1982) (A & F), Nicolet and Kennes (1989b) (N & K) and the current work with the line-by-line (LBL) calculations. Since we want to highlight the importance of seasonal and solar zenith angle variations on the parameterization of atmospheric transmittance and photodissociation rates, we compared these with the high spectral resolution calculation for typical summer, and winter atmospheres and a range of solar zenith angles. The absorption cross sections in intervals 16 and 17 do not contain many spectral features and are not included in N & K. In the comparison we have used the effective cross sections

obtained from our calculations: 7.9×10^{-24} and 7.31×10^{-24} cm², respectively. The Herzberg continuum absorption cross sections available at the time of the A & F work were considerably overestimated. We reduced their values for the effective cross sections in the intervals 13–17 by 40% to account for this. Although not rigorously correct, this approach is commonly taken in the modelling community (private communication Ch. Brühl).

Figure 7a–c shows typical calculations of O-production rates by the LBL model and the differences when using a parameterization. Differences from the LBL calculations can arise from: (1) inadequacies in the parameterizations and (2) the use of different spectroscopic data available at the time the parameterization was made. For the current parameterization differences are solely due to errors induced by using the parameterization, as it is directly based on our LBL calculations. Figure 7a shows the difference in O-production rates between the different parameterizations for overhead Sun and US Standard Atmosphere. Differences are thus solely due to different spectroscopic data. The current parameterization overpredicts O-production rates with a maximum error of about 2% at altitudes with more than 1% of the maximum O-production rate based on a 5-km altitude interval. For A & F this is 9% and for both N & K and WMO more than 12% in a given altitude range.

As we wanted to test the parameterizations' ability to deal with optical path variations, we chose to normalize all parameterizations to their O-production profile at overhead Sun and US Standard Atmosphere, thereby minimizing the effect of different spectroscopic data. This gives a fairer representation of the parameterizations' performance. The errors indicated in Fig. 7b, c are thus solely due to the inability to handle optical path variations.

Figure 7b is calculated for different solar zenith angles. O-production rates decrease and the level for maximum energy dissipation shifts upwards with increasing solar zenith angles. Clearly, the other tested parameterizations are over large altitude regions, unable to also deal with solar zenith angle variations. The magnitude of the errors increases with solar zenith angle. We chose not to include N & K at 85° since the results were physically unrealistic at these large solar zenith angles.

Figure 7c is calculated for overhead Sun in different typical high-latitude seasons, again although physically unrealistic at these latitudes, illustrating parameterization performance. The high-latitude summer mesopause temperatures imply an increased transmittance in this altitude region, resulting in increased O-production rates directly below compared to the standard atmosphere. The current parameterization induces the largest errors (a maximum of about 3%) for the typical high-latitude summer atmosphere due to the extremely low temperatures. A parameterization that is not more or less able to "catch" the temperature minimum can, as seen in Fig. 7b, c, introduce errors up to about 10% in this altitude range. Accounting for both error sources (parameterization and spectroscopic data) this can reach 15%. Note that differences in altitude regions with low O-production rates (for example, WMO below 40 km for 85°) are not important for the column O-production. Spectral intervals/altitude ranges

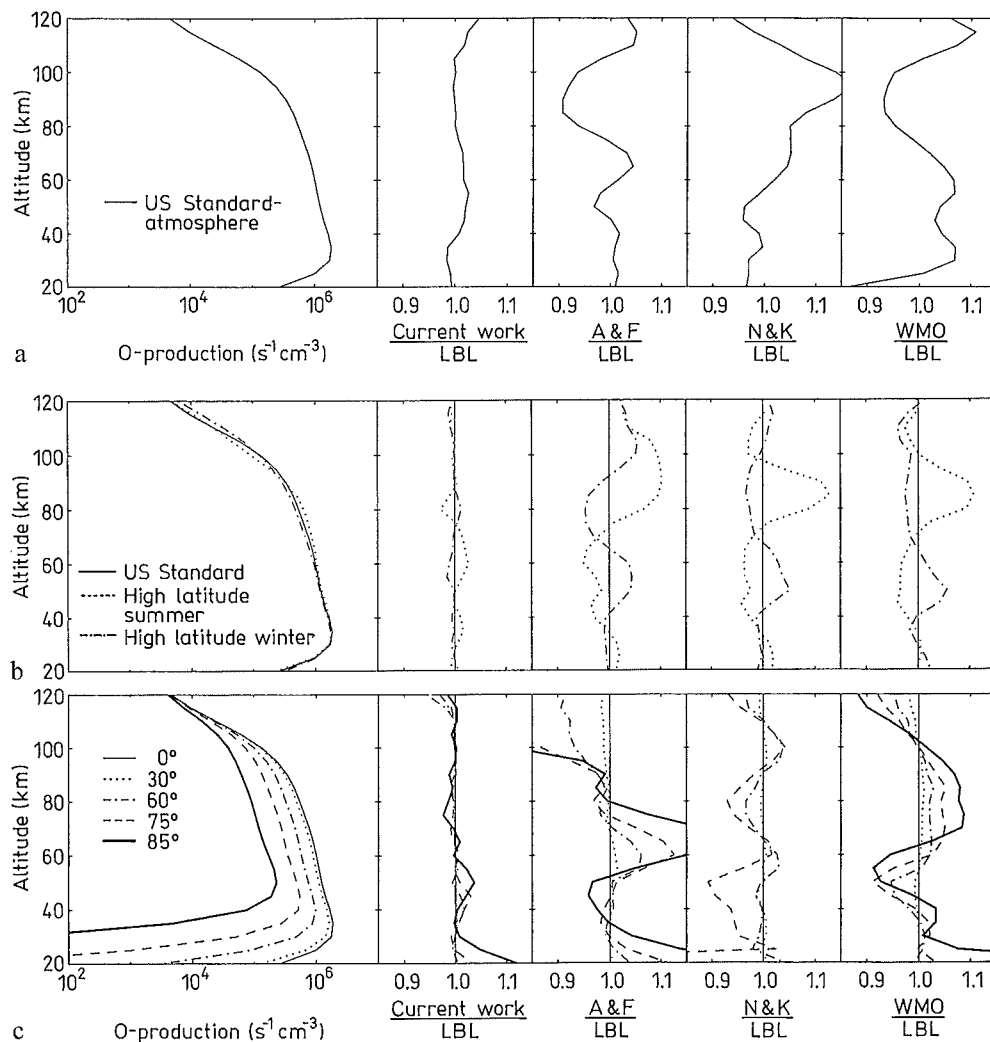


Fig. 7. **a** Left panel: O-production rates in the Schumann-Runge bands for overhead Sun from the line-by-line calculations in the US Standard Atmosphere. The four panels to the right show the difference from the LBL calculations the current work, Allen and Frederick, Nicolet and Kennes and WMO. **b** As **a** but for different solar zenith angles and now normalized to the respective parameterization performance for overhead Sun in the US Standard Atmosphere. **c** As **b** but for overhead Sun and typical high-latitude summer and winter atmospheres again normalized to the respective parameterization performance for overhead Sun in the US Standard Atmosphere

with less than 1% of the column O-production have therefore not been included in the error analysis.

Table 3a summarizes Fig. 7a–c. Both the errors in total column O-production rate and the maximum error in the O-production rate in a layer of 5 km at a given altitude, for those altitudes where the O-production rate was more than 5% of the column O-production rate, are given. Here we distinguish between the absolute differences between the LBL calculations and the respective parameterizations due to changes in the used spectroscopic data and due to parameterization errors. Column production rates are calculated to within a few percent by all three models for solar zenith angles less than 60°. A & F tend to give the best column production rate for solar zenith angles between 0° and 60°. All methods except for N & K at 75° tend to overpredict the column O-production rate. The maximum errors in a 5 km thick layer at a given altitude vary significantly between the different methods and increase generally with increasing solar zenith angle. They can be reduced significantly by using the current parameterization. It should be noted that diurnal photochemical models always come into the regime of solar zenith angles larger than 60° near sunset and sunrise at any

latitude. At high latitudes one can be in this regime during the whole model run depending on the season. This makes a correct parameterization at large zenith angles essential.

Table 3b is similar to Table 3a, but for transmittance. Transmittance for the current parameterization is calculated using Eq. (8), while both A & F and WMO are calculated according to their references. N & K parameterize the transmittance separately. For any altitude, season or solar zenith angle, the current work estimates transmittance to within 4% of the LBL calculations. At solar zenith angles larger than say 60° A & F, N & K and WMO tend to underestimate atmospheric transmittance by up to an order of magnitude difference at 85°.

The impact on the accuracy of the transmittance calculation including Eq. (8) has been tested with this work's parameterization for two layer thicknesses: 1 and 5 km. We compared the readily obtained transmittance and the O-production rates per altitude range from the parameterization with the detailed LBL calculations for both layer thicknesses with simple spatial averaging of the effective cross section and with Eq. (8). We found no significant increase in the error introduced by the current parameterization when the layer thickness was increased

Table 3a. Comparison of the O-production calculated by line by line calculations and three parameterization methods, for three cases: US Standard-, summer-, and winter-atmospheres. The values are the percentage error in total column O-production and maximum difference in O-production in an altitude interval of $\Delta z = 1$ km, in the whole Schumann-Runge band, from the line-by-line calculations

	Current work		Allen & Frederick		Nicolet & Kennes		WMO	
	Total	Max	Total	Max	Total	Max	Total	Max
χ	US Standard							
0	0.555	2.684	-0.323	4.394	0.438	5.047	3.165	7.078
30	0.620	2.767	-0.197	4.573	0.405	-5.180	3.266	7.246
60	0.839	2.638	0.437	5.552	0.122	-4.448	3.116	9.140
75	1.026	2.995	1.402	9.480	-4.465	-11.590	3.116	11.550
85	1.061	2.784	2.799	20.180	> 100	> 100	3.477	15.730
χ	Summer							
0	1.318	4.393	-0.719	-4.749	0.008	-8.430	2.807	7.545
30	1.403	4.619	-0.570	-3.903	0.129	-7.310	2.833	7.118
60	1.677	4.936	0.199	2.649	-0.370	-7.802	2.890	7.305
75	1.876	5.456	1.384	6.150	-5.014	-12.900	3.158	13.960
85	1.882	5.510	3.112	16.950	> 100	> 100	3.823	22.520
χ	Winter							
0	0.221	-2.124	0.010	5.298	0.610	5.717	3.809	10.050
30	0.281	2.033	0.149	5.477	0.921	5.421	3.692	10.700
60	0.465	2.087	0.858	6.907	0.731	-4.425	3.768	12.070
75	0.587	2.061	1.915	11.790	-4.031	-10.010	3.784	13.610
85	0.603	2.362	3.304	22.070	> 100	> 100	3.923	15.500

Table 3b. As 3a but for transmittance. Only the maximum errors in the altitude range between 20 and 120 km are given

	Current work		Allen & Frederick		Nicolet & Kennes		WMO	
	Total	Max	Total	Max	Total	Max	Total	Max
χ	US Standard							
0	1.092	2.772	-3.703	-27.820				
30	1.141	2.668	-4.130	-31.410				
60	-1.397	-2.740	-6.649	-46.870				
75	-1.993	-8.658	-16.250	-67.360				
85	-2.712	-26.410	> 100	-93.040				
χ	Summer							
0	1.998	3.828	-4.342	-27.610				
30	2.048	3.554	-4.191	-31.300				
60	-2.405	2.204	-4.543	-47.380				
75	-2.676	-8.299	-49.700	-68.260				
85	-3.972	-28.420	> 100	-93.710				
χ	Winter							
0	-0.761	2.063	-4.830	-25.790				
30	-0.690	2.203	-5.194	-28.720				
60	-1.194	-3.518	-7.507	-43.420				
75	-1.864	-8.863	-16.280	-63.780				
85	-1.978	-25.370	-65.650	-91.040				

from 1 to 5 km when Eq. (8) was employed whereas the simple spatial average cross-section approach alone caused errors up to 15% in transmission at the lower boundary for the $\Delta z = 5$ km case. It should of course be

noted that increasing the altitude step eventually will cause problems in regions with sharp temperature gradients and should be avoided.

A comparison between the computational speed of the tested parameterizations has been made. The decrease in computational speed applying this study's parameterizing with use of Eq. (8) is seen in Fig. 6 as a function of the number of coefficients used.

4.2 Photochemical modelling results

From Fig. 7a–c it is clear that the different parameterizations produce different quantities of oxygen atoms. There are, as seen from Table 3b, also large effects on the transmittance depending on which parameterization is used. Whereas the O-production rates can vary in sign compared to our LBL calculation we calculate increased transmittance at solar zenith angles larger than 60° compared with both N & K and WMO. This means that under these conditions a number of minor species will be photodissociated at rates that are larger than previously estimated. The impact of this increased photodissociation rate will depend on the species in question. For example, in the case of water vapour, which has a dissociation cross section decreasing by several orders of magnitude in the Schumann-Runge band region, this means that a small increase in insolation at the short-wavelength end of the SR-Band region can lead to significantly increased photodissociation rates. The largest increase in transmittance actually occurs at the short-wavelength end. It should be

noted that using constant H_2O absorption cross-section in 500 cm^{-1} intervals, as applied in all our comparative model runs, is erroneous (Siskind *et al.*, 1994). However, many currently implemented models use this approach and the results presented are relevant to them. Additionally, they serve to illustrate the effect on a minor species whose photolysis rate depends on the transmittance in the Schumann-Runge bands.

Above 70 km, H_2O is mainly dissociated by $Ly-\alpha$, but dissociation in the Schumann-Runge bands increases in importance with decreasing altitude so, as seen in Fig. 8a–c, the changes in odd-hydrogen production between the different parameterizations are mainly below this altitude. The final effect of the new parameterization on odd oxygen will be the combined effect of increased or decreased O-production rates depending on the altitude

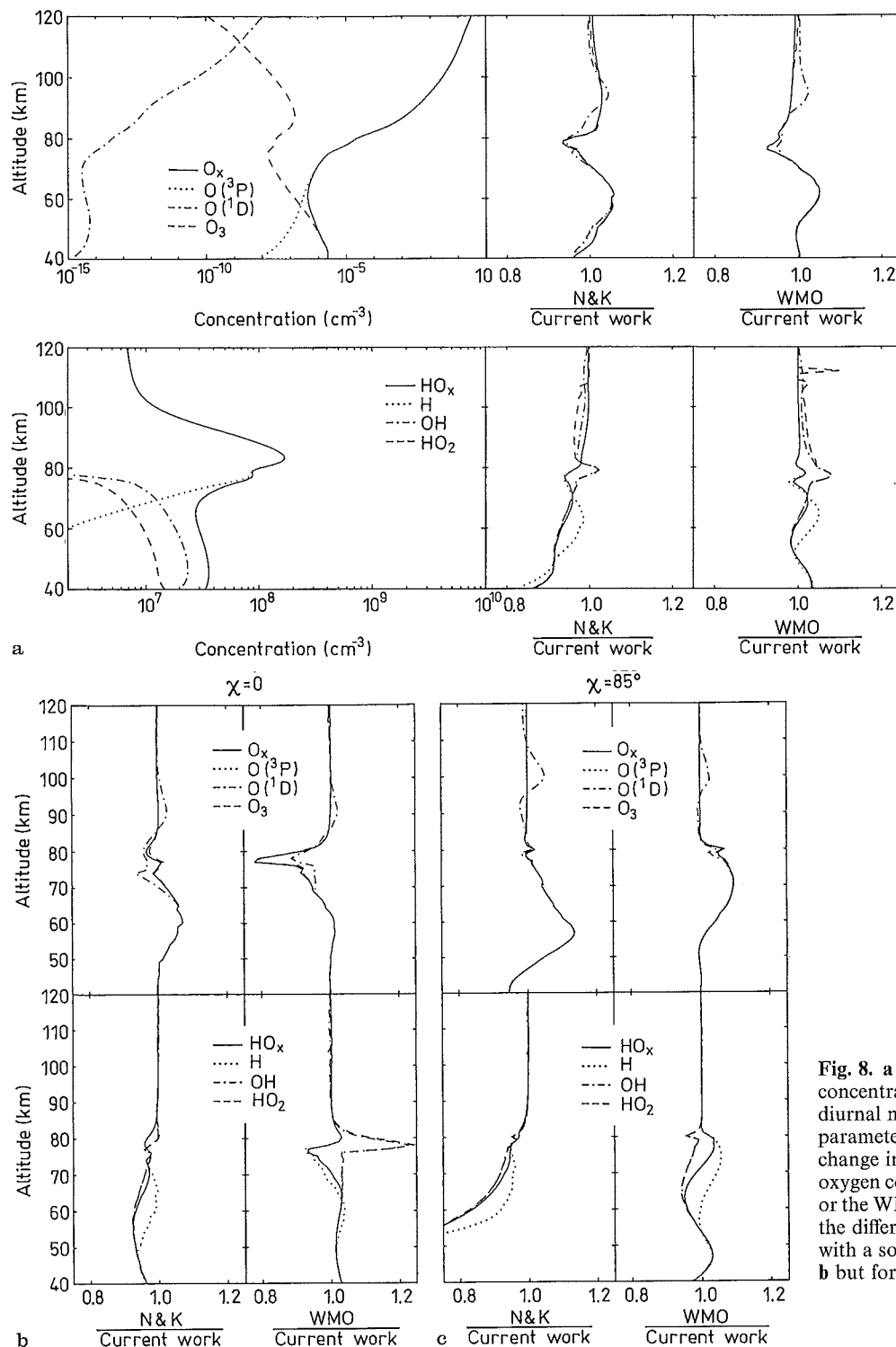


Fig. 8. a Odd-hydrogen and odd-oxygen concentrations from a 1D photochemical diurnal model using the current parameterization. The right panels show the change in percent in odd-hydrogen and odd-oxygen concentrations when using either N&K or the WMO parameterization. b As a but only the differences from a steady-state model run with a solar zenith angle of 0° are shown. c As b but for a solar zenith angle of 85°

range and optical path variations and a general increase in transmittance leading, among others, to increased odd-hydrogen production. Fig. 8a–c illustrates the resulting effect on odd-hydrogen and odd-oxygen concentrations for overhead Sun and 85°. Odd-hydrogen production below about 80 km is significantly lower when using N & K than that of the current work. The effect is a clear increase of odd-oxygen concentrations up to the mesopause. Generally, deviations between O_x concentrations are less than 10% for overhead Sun and 10–20% near sunset and sunrise. Deviations between HO_x concentrations can be up to 30–40% at the lower boundary at sunset and sunrise and are mainly due to differences in transmittance parameterizations.

5 Conclusions

Comparison between the different parameterizations and direct LBL calculations show that the current parameterization is able to handle the large optical path fluctuations associated with solar zenith angle and seasonal variations. The other parameterizations tested tend to perform significantly worse. Specifically when calculating transmittance at large solar zenith angles the current parameterization shows superior performance. Computational effort increases compared to the simplest parameterization and depends on the number of coefficients used in the algorithm but should not be a direct problem since the actual transmittance and photodissociation calculations in this wavelength region are not the most time-consuming procedures in most photochemical models.

Chemical modelling shows that ozone chemistry is sensitive to the choice of the parameterization method. Odd-oxygen concentrations are modified by the competition between increased O-production rates and the increased photodissociation rate of water vapour. The first tends to increase odd-oxygen concentrations, whereas the second tends to decrease it. The overall effect modifies odd-oxygen concentrations at selected altitudes by up to 15% at fixed solar zenith angles and up to 10% at noon in the case of full diurnal calculations. Deviations at larger solar zenith angles around sunset, sunrise or at high latitudes, can be up to 20% at selected altitudes.

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